

Explainable AI-Enhanced Energy Forecasting Using LightGBM with SHAP and LIME Interpretability

K. Jnana Varshitha
Department of ECE

Amrita school of Engineering
Amrita Vishwa Vidyapeetham
Chennai, India
killivarshitha@gmail.com

L. Pavan Kumar
Department of ECE,

Amrita school of Engineering
Amrita Vishwa Vidyapeetham
Chennai, India
lalampavan888@gmail.com

Pavan Manellore
Department of ECE

Amrita school of Engineering
Amrita Vishwa Vidyapeetham
Chennai, India
pavanmanellore@gmail.com

B. Devanathan
Department of ECE

Amrita school of Engineering
Amrita Vishwa Vidyapeetham
Chennai, India
b_devanathan@ch.amrita.edu

Lakshmanan S A
Department of ECE,

Amrita school of Engineering
Amrita Vishwa Vidyapeetham
Chennai, India
sa_lakshman@ch.amrita.edu

N. Krishna Prakash

Department of Energy and Climate Change
School of Environment, Resources and Development
Asian Institute of Technology
Pathum Thani, Thailand
krishna.prakash.natarajan@ait.ac.th

Abstract— Accurate prediction of photovoltaic generation, wind generation, and electric demand is crucial for grid reliability, economic dispatch, and the integration of renewable resources across different operational timeframes. This study proposes a forecasting framework that evaluates multiple machine learning models, including XGBoost, Random Forest, NGBoost, CatBoost, SVR, and KNN, with LightGBM identified as the most effective learner when applied to 5-minute data aggregated into daily intervals. To enhance transparency, Explainable AI techniques are integrated: SHAP quantifies global and local feature contributions, highlighting the influence of lagged signals, while LIME provides case-specific explanations for individual forecasts. The methodology employs chronological data splits with train-only scaling to preserve the integrity of time-series learning and uses R^2 and RMSE metrics for evaluation. Results indicate that forecasting performance varies across datasets, underscoring the need for customized models. Comparative analysis confirms that LightGBM achieves the best trade-off between accuracy, stability, and interpretability. By combining robust forecasting with explainable insights, the proposed framework supports AI-enhanced energy management, enabling situational awareness, informed decision-making, and scalable deployment in renewable-rich power systems.

Index Terms—Forecasting, LightGBM, Explainability, SHAP, LIME, Photovoltaic, Wind.

I. INTRODUCTION

Reliable forecasting of photovoltaic generation, wind generation, and electric demand is essential to balance supply and demand, reduce operational costs, and enable higher renewable penetration. This work sets out to design and validate a transparent forecasting framework built on rigorous time-series practices, including chronological data partitioning, train-only scaling, and comprehensive evaluation using Performance Metrics. The study focuses on transforming high-frequency measurements into operationally meaningful resolutions, engineering lagged features to capture temporal dependencies,

and systematically assessing performance heterogeneity. To address the black-box challenge that limits trust and adoption in operational settings, the framework integrates explainability techniques to reveal global drivers and provide instance-level rationales for predictions, clarifying how lagged signals conditions shape outcomes. The objective is to deliver accurate, interpretable forecasts that support situational awareness, risk-aware decision-making, and scalable deployment in renewable-rich systems without presupposing specific algorithms, thereby emphasizing methodological rigor, interpretability, and real-world applicability.

A. Related Work

Presents an adaptive renewable energy forecasting methodology utilizing a data-driven PCA-Transformer architecture to improve prediction accuracy through dimensionality reduction and attention mechanisms [1]. Develops an explainable AI framework combining XGBoost with SHAP and LIME interpretability techniques for multi-scale household energy consumption forecasting applications [2]. Proposes an attention-driven hybrid ensemble approach integrated with Bayesian optimization for accurate renewable energy forecasting specifically in Jeju Island's power system [3]. Introduces feature driven methodologies to improve both renewable energy forecasting accuracy and energy trading performance in power markets [4]. Explores prediction markets as a novel approach for probabilistic forecasting of renewable energy sources to better handle uncertainty quantification [5]. Provides a comprehensive survey examining multimodel learning techniques applied to time series forecasting problems in renewable energy systems [6]. Optimizes power system reliability and carbon emissions through a fuzzy unit commitment model that incorporates renewable energy sources, load forecast uncertainties, electric vehicle charging,

and energy storage systems [7]. These studies collectively advance renewable energy forecasting through machine learning, explainable AI, ensemble methods, and optimization techniques while addressing practical implementation challenges in modern power systems [8]. Proposes an energy consumption prediction model utilizing LightGBM empowered with explainable artificial intelligence (XAI) to enhance forecasting accuracy and model interpretability for households and businesses [9]. Develops an explainable artificial intelligence-based photovoltaic energy management framework specifically designed for smart city applications to enhance solar energy utilization and transparency [10]. Presents a cloud-integrated virtual framework utilizing LSTM networks for solar forecasting and residential energy management to optimize household energy consumption patterns [11]. Utilizes explainable artificial intelligence tools to gain insight into solar photovoltaic power generation forecasting and improve model interpretability [12]. Develops an explainable AI-driven dew point forecasting model using attention-based temporal convolutional networks for enhanced weather prediction accuracy [13]. Reveals the effects of data heterogeneity in federated learning regression models for short-term solar power forecasting applications [14]. Proposes a probabilistic LSTM-autoencoder based hour-ahead solar power forecasting model for intra-day electricity market participation in Poland [15]. The structure of this paper is organized as follows: Section II presents the methodology in this study. Section III reports the forecasting results and provides a discussion of the key findings. Finally, Section IV concludes the paper by summarizing the major contributions and insights.

II. METHODOLOGY

This methodology aims to generate precise and comprehensible forecasts for photovoltaic generation, wind generation, and electric demand over various time horizons, all while maintaining the integrity of time-series data and ensuring operational realism. This approach converts high-frequency data into operationally relevant time frames, establishes temporal dependencies through lagged features, employs stringent chronological assessment with train-only scaling, and integrates model explainability to link predictions to influential factors, facilitating both system-level planning and immediate operational decision-making. The proposed framework is depicted in Figure 1.

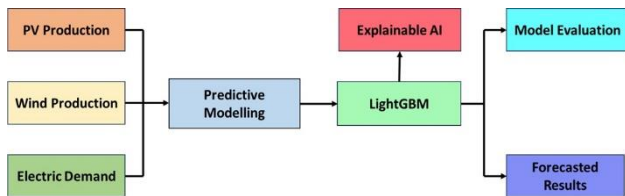


Fig. 1: Proposed Framework.

A. Dataset

The dataset was collected from sources provided by the California Independent System Operator and the National

Renewable Energy Laboratory, sampled uniformly at five-minute intervals, and subsequently subjected to a thorough cleaning process to produce three synchronized time series representing solar production, wind production, and electricity demand. It consists of 12 features, timestamped date index; a season code indicating Winter, Spring, Summer, or Autumn with values ranging from 1 to 4; a weekday indicator represented as 0 to 6 for Monday through Sunday; three measures of irradiance diffuse horizontal (DHI, W/m²), direct normal (DNI, W/m²), and global horizontal (GHI, W/m²); meteorological variables such as wind speed (m/s), relative humidity (%), and ambient temperature (degrees); along with three target variables measured in megawatts that reflect solar energy production, wind energy production, and total electricity demand. All variables are synchronized in time at five-minute intervals.

B. Forecasting Pipeline

The forecasting pipeline is executed independently for each horizon by resampling the data, adding lagged features, and splitting chronologically into train and test to preserve temporal order. Feature scaling is fit only on the training predictors and applied to both sets to avoid leakage, while targets are min-max scaled on the training window and inverse-transformed after prediction to recover MW for evaluation. Seven machine learning models were trained on the scaled training matrix and used to generate test predictions; these are inverse-transformed and assessed using R², MSE, RMSE, and MAE and based on these performance metrics LightGBM performs well. Global interpretability is provided with SHAP summary plots to quantify each input's contribution distribution across the test set, clarifying the influence of recent lags, while local interpretability is obtained by explaining an exemplar test instance to show the dominant drivers of a specific forecast.

C. LightGBM model

LightGBM is a gradient boosting framework that builds an additive ensemble of decision trees to minimize a chosen loss via stage-wise functional gradient descent, offering efficiency and strong accuracy for tabular time-series with lagged and exogenous inputs in multi-horizon forecasting. In regression, predictions are modelled as

$$\hat{y} = F_m(x_t) = \sum_{m=1}^M f_m(x_t)$$

where each f_m is a regression tree added to correct residual errors of the current model F_{m-1} through the update

$$F_m(x) = F_{m-1}(x) + \gamma f_m(x), \quad \eta \in (0,1]$$

with learning rate η . Using a second-order Taylor approximation of the loss ℓ , LightGBM computes per-sample gradient and Hessian

$$g_t = \frac{\partial \ell}{\partial y} \Big|_{y = F_{m-1}(x_t)} \quad h_t = \frac{\partial^2 \ell}{\partial y^2} \Big|_{y = F_{m-1}(x_t)}$$

and for squared loss one has

$$g_t = F_{m-1}(x_t) - y_t \quad h_t = 1$$

For a leaf containing index set I , the optimal leaf weight with L2 regularization λ is

$$\omega^* = \frac{-\sum_{t \in I} g_t}{\sum_{t \in I} h_t + \lambda}$$

with leaf score

$$S(I) = \frac{-1}{2} \frac{\sum_{t \in I} g_t^2}{\sum_{t \in I} h_t + \lambda}$$

For a candidate split $I \rightarrow (I_L, I_R)$, the regularized gain is

$$Gain = S(I_L) + S(I_R) - S(I) - \tau$$

which LightGBM evaluates efficiently via histogram-based binning and expands leaves in a best-first manner to reduce complexity while preserving strong fit.

III. RESULTS AND DISCUSSION

The findings are organized according to operational horizon and target, followed by analyses of explainability utilizing SHAP and LIME to provide context for model behavior. In each scenario, models are developed using standardized predictors and min-max-scaled targets, with an inverse transformation applied during inference to present performance in megawatts. Empirical evidence shows that the LightGBM-based pipeline achieves high R^2 and low RMSE for photovoltaic (PV), wind, and demand across various test windows. Additionally, SHAP attributions consistently identify recent lag features as crucial factors influencing predictions.

A. Daily Forecasting

Daily models exhibit a strong correlation between the predicted and observed series for photovoltaic (PV), wind energy, and demand, effectively capturing day-to-day fluctuations while maintaining overarching trends throughout the testing period. The errors, once inverse-transformed, remain low in terms of absolute megawatts, and the R^2 values as shown in Tables I indicate significant explanatory power derived from recency effects represented by lagged features,

further supported by external weather and irradiance signals that adjust forecast level in response to varying conditions as shown in Figures 2, 3, 4.

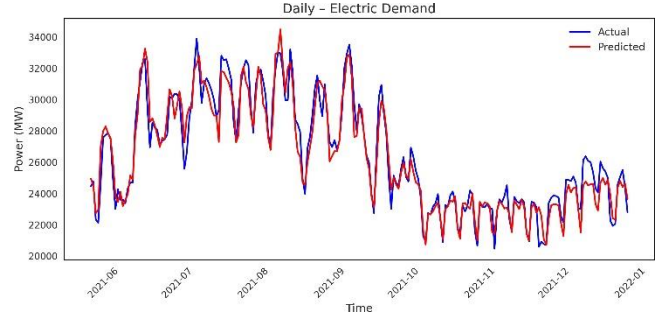


Fig. 2: PV Production Forecasting.

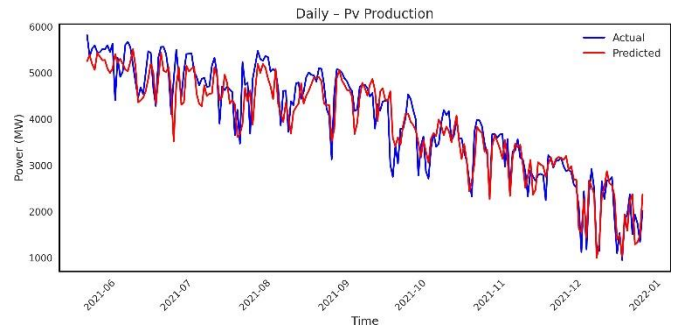


Fig. 3: Wind Energy Production Forecasting.

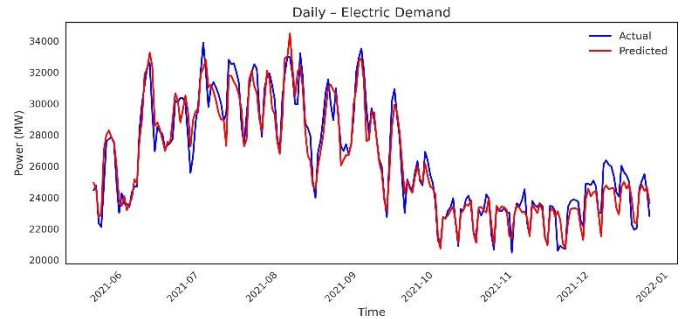


Fig. 4: Electric Demand Forecasting.

B. SHAP

SHAP decomposes each prediction into additive feature attributions that sum to the difference between the prediction and a baseline, providing transparent accounting of inputs' effects. In contrast, PV production Figure 5 experiences a

TABLE I: Comparison of Performance Metrics for PV Production, Wind Energy Production, and Electric Demand.

Model	PV Production				Wind Energy Production				Electric Demand			
	MAE	MSE	RMSE	R ²	MAE	MSE	RMSE	R ²	MAE	MSE	RMSE	R ²
LGBM	0.0062	0.0052	0.0121	0.9985	0.0243	0.0237	0.0538	0.9260	0.0341	0.0020	0.0340	0.9386
XGBoost	0.0584	0.0053	0.0729	0.8828	0.1295	0.0259	0.1610	0.4811	0.0347	0.0020	0.0448	0.9230
Random Forest	0.0586	0.0054	0.0736	0.8806	0.1260	0.0254	0.1592	0.4920	0.0355	0.0023	0.0478	0.9306
KNN	0.0862	0.0108	0.1038	0.7625	0.1293	0.0295	0.1718	0.4087	0.0594	0.0057	0.0757	0.8259
CatBoost	0.0609	0.0054	0.0736	0.8808	0.1287	0.0248	0.1573	0.5041	0.0291	0.0015	0.0386	0.9347
NGBoost	0.0728	0.0081	0.0902	0.8207	0.1314	0.0267	0.1635	0.4646	0.0358	0.0022	0.0470	0.9330
SVR	0.1008	0.0147	0.1214	0.6754	0.1467	0.0322	0.1793	0.3557	0.0603	0.0058	0.0759	0.8252

more rapid decay: autocorrelation decreases from 1.0 to approximately 0.4 at lag-1 and approaches zero by lag-2, this limited memory suggests that PV forecasts should heavily prioritize the most recent data while disregarding observations older than two steps. Wind production as shown in Figure 6 finds itself in an intermediate position: correlations remain around 0.2 to 0.25 for the initial two lags before diminishing quickly. In summary, the graphs quantify the extent to which each source retains memory of its past, which directly informs the optimal lookback period for forecasting models. Electric demand as shown in Figure 7 exhibits the most robust and enduring persistence lags 1-3 consistently remain above 0.3, nearly linear decline indicates a highly predictable process characterized by inertia, influenced by human habits and thermal mass.

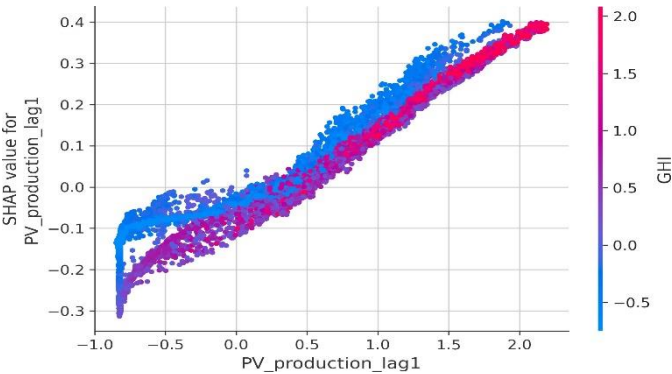


Fig. 5: SHAP Value plot for PV Production.

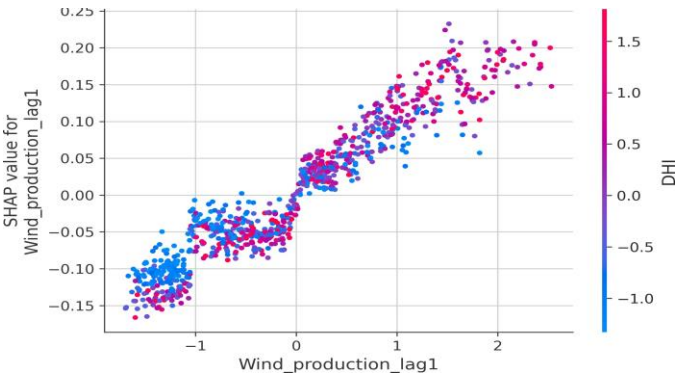


Fig. 6: SHAP Value plot for Wind Energy Production.

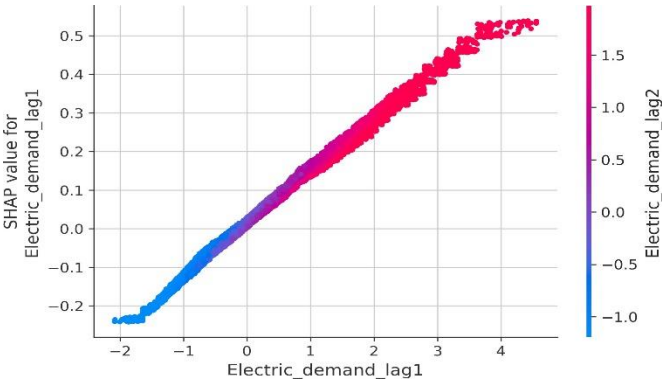


Fig. 7: SHAP Value plot for Electric Demand.

C. LIME

LIME provides local, human-readable explanations by fitting sparse surrogates around specific test instances, quantifying which features most directly drove a given

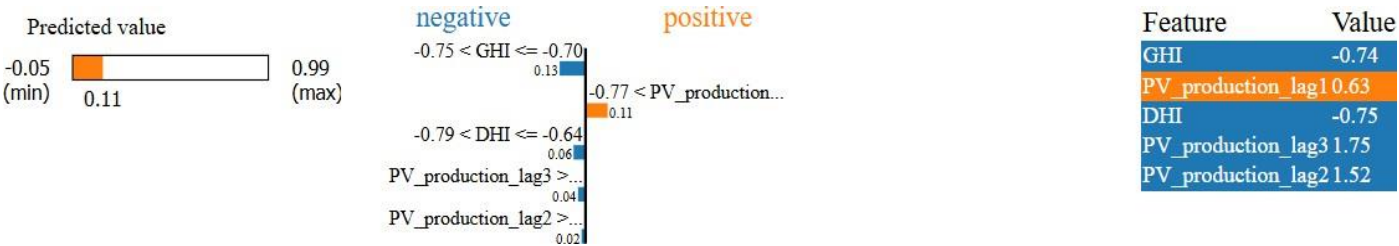


Fig. 8: LIME Explainer Plot for PV Production.

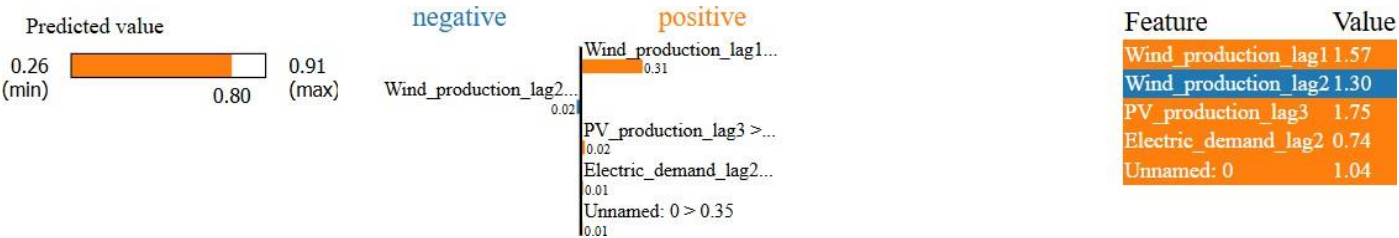


Fig. 9: LIME Explainer Plot for Wind Energy Production.

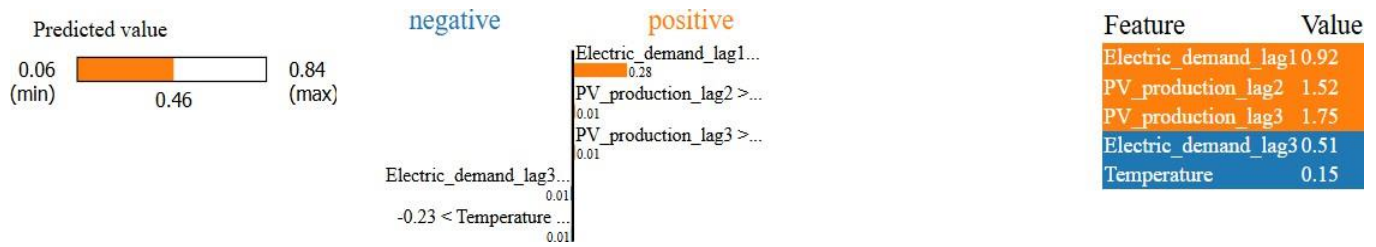


Fig. 10: LIME Explainer Plot for Electric Demand.

forecast at a particular time. Instance-level analyses highlight the relative roles of recent lags, irradiance, wind speed, humidity, and calendar effects in producing spikes or dips, supporting operator-facing justifications and targeted incident reviews where errors are largest or atypical. PV production Figure 8 is inversely related to low GHI/DHI bins (-0.74, -0.75), indicating that reductions in irradiance during cloudy conditions lead to a sharp decline in output; concurrently, its own lag-1 (0.63) and lag-3 (1.75) provide positive memory. Wind production Figure 9 exhibits significant autocorrelation at lag-1 (1.57) and lag-2 (1.30), with minor positive contributions from PV lag-3 and electric demand lag-2. Electric demand Figure 10 is predominantly influenced by its own 1-hour lag (0.92) and temperature have minimal impact on the prediction, thereby affirming that recent demand history is adequate for short-term forecasting, thus the forecast penalizes adverse conditions but favours recent PV performance.

IV. CONCLUSION

The findings validate that the suggested framework produces precise and interpretable predictions for photovoltaic generation, wind generation, and electricity demand. The LightGBM based pipeline consistently surpasses alternative learners in daily evaluations regarding both fit and error magnitude. In terms of PV, wind and demand, metrics exhibit high R^2 values alongside low RMSE and MAE, which signifies a strong correlation between predicted and actual trajectories. The analysis of explainability further support these conclusions SHAP consistently links forecast variations to recent lags, while LIME offers case-specific justifications that clarify the reasons behind particular spikes. Collectively, these results validate the methodological integration of robust forecasting with reliable explanations, risk-aware operations, and scalable implementation in power systems rich in renewable resources.

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